

Day 04

Run and record output for at least 6 commands :

Process commands (ps, top, pgrep)

ps -aux | head -n 5 = It displays **the top 5 lines** (including the header) from the list of all running processes

```
ubuntu@ip-172-31-23-127:~$ ps -aux | head -n 5
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.1  1.4 22768 13900 ?        Ss   16:05   0:01 /sbin/init
root             2  0.0  0.0      0     0 ?        S    16:05   0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    16:05   0:00 [pool_workqueue_release]
root             4  0.0  0.0      0     0 ?        I<   16:05   0:00 [kworker/R-rcu_gp]
```

top | head -n 10 = It shows the **first 10 lines of live system and process information** from top

```
top - 16:31:59 up 26 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 117 total, 1 running, 112 sleeping, 4 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 914.2 total, 104.0 free, 393.5 used, 575.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 520.7 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
 1257 ubuntu    20   0   15252   7428   5216 S   10.0   0.8   0:00.71 sshd
    1 root      20   0   22768  13900   9584 S    0.0   1.5   0:01.44 systemd
    2 root      20   0      0      0      0 S    0.0   0.0   0:00.00 kthreadd
```

pgrep -l docker = Finds processes whose name **matches nginx** and shows the **PID with process name**.

```
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$ pgrep -l docker
811 dockerd
ubuntu@ip-172-31-23-127:~$
```

Service commands (systemctl status, systemctl list-units, etc)

systemctl status nginx = It shows the **current status of the Nginx service**

```

ubuntu@ip-172-31-23-127:~$ systemctl status nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run 'systemctl daemon-reload' to reload units.
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-02-07 16:05:15 UTC; 40min ago
     Docs: man:nginx(8)
   Main PID: 601 (nginx)
    Tasks: 3 (limit: 1017)
  Memory: 3.7M (peak: 4.1M)
     CPU: 23ms
    CGroup: /system.slice/nginx.service
            └─601 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─602 "nginx: worker process"
                └─603 "nginx: worker process"

Feb 07 16:05:15 ip-172-31-23-127 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Feb 07 16:05:15 ip-172-31-23-127 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-23-127:~$

```

systemctl list-units --type=service --state=running | head = It shows the **first few running services** on the system.

```

ubuntu@ip-172-31-23-127:~$ systemctl list-units --type=service --state=running | head
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
acpid.service                       loaded active running ACPI event daemon
chrony.service                       loaded active running chrony, an NTP client/server
containerd.service                  loaded active running containerd container runtime
cron.service                         loaded active running Regular background program processing daemon
dbus.service                         loaded active running D-Bus System Message Bus
docker.service                       loaded active running Docker Application Container Engine
getty@tty1.service                  loaded active running Getty on tty1
irqbalance.service                  loaded active running irqbalance daemon
ModemManager.service                 loaded active running Modem Manager

```

Log commands (journalctl -u <service>, tail -n 50, etc.)

journalctl -u docker | head -n 10 = It shows the **first 10 log entries** for the **Docker service** from the system logs.

```

ubuntu@ip-172-31-23-127:~$ journalctl -u docker | head -n 10
Feb 03 07:01:45 ip-172-31-23-127 systemd[1]: Starting docker.service - Docker Application Container Engine...
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.111134987Z" level=info msg="Starting up"
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.113188912Z" level=info msg="OTEL tracing is not configured, using no-op tracer provider"
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.113570572Z" level=info msg="detected 127.0.0.53 nameserver, assuming systemd-resolved, so using resolv.conf: /run/systemd/resolve/resolv.conf"
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.164646576Z" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock timeout=1m5s
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.220886312Z" level=info msg="Loading containers: start."
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.752965049Z" level=info msg="Loading containers: done."
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.774124162Z" level=info msg="Docker daemon" commit="28.2.2-0ubuntu1~24.04.1" containerd-snapshotter=false storage-driver=overlay2 vers
v28.2.2
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.774236055Z" level=info msg="Initializing buildkit"
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.779793740Z" level=warning msg="CDI setup error /etc/cdi: failed to monitor for changes: no such file or directory"
ubuntu@ip-172-31-23-127:~$

```

journalctl -u docker | grep -i error | head = Shows the first 10 Docker service log entries that contain the word "error".

```

ubuntu@ip-172-31-23-127:~$ journalctl -u docker | grep -i error | head
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.779793740Z" level=warning msg="CDI setup error /etc/cdi: failed to monitor for changes: no such file or directory"
Feb 03 07:01:45 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T07:01:45.779843201Z" level=warning msg="CDI setup error /var/run/cdi: failed to monitor for changes: no such file or directory"
Feb 03 10:54:52 ip-172-31-23-127 dockerd[2634]: time="2026-02-03T10:54:52.582289691Z" level=info msg="stopping event stream following graceful shutdown" errors=<nil> module=libcontainerd namespace=moby
Feb 07 03:42:57 ip-172-31-23-127 dockerd[862]: time="2026-02-07T03:42:57.442747692Z" level=warning msg="Error (Unable to complete atomic operation, key modified) deleting object [endpoint_count 3899a30bf2dff9615abb09b0b226faf7282506c2798f28920a093d1f370e458b], retrying..."
Feb 07 03:42:57 ip-172-31-23-127 dockerd[862]: time="2026-02-07T03:42:57.535284115Z" level=warning msg="CDI setup error /var/run/cdi: failed to monitor for changes: no such file or directory"
Feb 07 03:42:57 ip-172-31-23-127 dockerd[862]: time="2026-02-07T03:42:57.535284115Z" level=warning msg="CDI setup error /etc/cdi: failed to monitor for changes: no such file or directory"
Feb 07 12:47:34 ip-172-31-23-127 dockerd[862]: time="2026-02-07T12:47:34.424738827Z" level=info msg="stopping event stream following graceful shutdown" errors=<nil> module=libcontainerd namespace=moby
Feb 07 12:47:34 ip-172-31-23-127 dockerd[862]: time="2026-02-07T12:47:34.426005896Z" level=info msg="stopping event stream following graceful shutdown" errors="context canceled" module=libcontainerd namespace=plugins.moby
Feb 07 16:05:17 ip-172-31-23-127 dockerd[811]: time="2026-02-07T16:05:17.645929962Z" level=warning msg="Error (Unable to complete atomic operation, key modified) deleting object [endpoint_count 73cd4467bb07f33796af0ba395a957d905d4cccae61868093ab9f9d8e64a52], retrying..."
Feb 07 16:05:17 ip-172-31-23-127 dockerd[811]: time="2026-02-07T16:05:17.728953277Z" level=warning msg="CDI setup error /etc/cdi: failed to monitor for changes: no such file or directory"
ubuntu@ip-172-31-23-127:~$

```

Service for Inspection commands (journalctl -u <service>, tail -n 50, etc.)

systemctl status nginx

sudo systemctl stop nginx

curl http://localhost

systemctl status nginx

journalctl -u nginx

sudo systemctl start nginx

systemctl status nginx

curl http://localhost

```
ubuntu@ip-172-31-23-127: ~$ sudo systemctl stop docker
Warning: The unit file, source configuration file or drop-ins of docker.service changed on disk. Run 'systemctl daemon-reload' to reload units.
Stopping 'docker.service', but its triggering units are still active:
docker.socket
ubuntu@ip-172-31-23-127:~$ sudo systemctl status docker
Warning: The unit file, source configuration file or drop-ins of docker.service changed on disk. Run 'systemctl daemon-reload' to reload units.
o docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: inactive (dead) since Sat 2026-02-07 17:35:06 UTC; 1min 29s ago
 Duration: 1h 29min 48.906s
 TriggeredBy: ● docker.socket
 Docs: https://docs.docker.com
 Process: 811 ExecStart=/usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock (code=exited, status=0/SUCCESS)
 Main PID: 811 (code=exited, status=0/SUCCESS)
 CPU: 1.000s

Feb 07 16:05:17 ip-172-31-23-127 dockerd[811]: time="2026-02-07T16:05:17.745036684Z" level=info msg="Completed buildkit initialization"
Feb 07 16:05:17 ip-172-31-23-127 dockerd[811]: time="2026-02-07T16:05:17.754947737Z" level=info msg="Daemon has completed initialization"
Feb 07 16:05:17 ip-172-31-23-127 dockerd[811]: time="2026-02-07T16:05:17.755164798Z" level=info msg="API listen on /run/docker.sock"
Feb 07 16:05:17 ip-172-31-23-127 systemd[1]: Started docker.service - Docker Application Container Engine.
Feb 07 17:35:06 ip-172-31-23-127 dockerd[811]: time="2026-02-07T17:35:06.662062056Z" level=info msg="Processing signal 'terminated'"
```

```
Feb 07 16:05:17 ip-172-31-23-127 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-23-127:~$ sudo systemctl stop nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run 'systemctl daemon-reload' to reload units.
ubuntu@ip-172-31-23-127:~$ curl http://localhost
curl: (7) Failed to connect to localhost port 80 after 0 ms: Couldn't connect to server
ubuntu@ip-172-31-23-127:~$ systemctl status nginx
```

```
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$ journalctl -u nginx
Feb 03 06:59:42 ip-172-31-23-127 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Feb 03 06:59:42 ip-172-31-23-127 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
Feb 03 07:07:43 ip-172-31-23-127 systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
Feb 03 07:07:43 ip-172-31-23-127 systemd[1]: nginx.service: Deactivated successfully.
Feb 03 07:07:43 ip-172-31-23-127 systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
Feb 03 07:08:33 ip-172-31-23-127 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Feb 03 07:08:33 ip-172-31-23-127 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
Feb 03 10:54:52 ip-172-31-23-127 systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
```

```
ubuntu@ip-172-31-23-127:~$
ubuntu@ip-172-31-23-127:~$ sudo systemctl start nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service changed on disk. Run 'systemctl daemon-reload' to reload units.
ubuntu@ip-172-31-23-127:~$ curl http://localhost
<img alt="nginx logo" data-bbox="114 765 190 768"/>
```

```
warning: the default 'type', source configuration file or drop-ins of nginx
ubuntu@ip-172-31-23-127:~$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
```

```
Feb 07 17:41:06 ip-172-31-23-127 systemd[1]: Stopping nginx.service - A high performance web server and a reverse proxy server...
Feb 07 17:41:06 ip-172-31-23-127 systemd[1]: nginx.service: Deactivated successfully.
Feb 07 17:41:06 ip-172-31-23-127 systemd[1]: Stopped nginx.service - A high performance web server and a reverse proxy server.
Feb 07 17:44:02 ip-172-31-23-127 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Feb 07 17:44:02 ip-172-31-23-127 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-23-127:~$
```

I first verified that the NGINX service was running, then intentionally stopped it to simulate a failure and confirmed the service outage.

After inspecting the service status and logs, I restarted NGINX and verified that the web service was restored successfully.